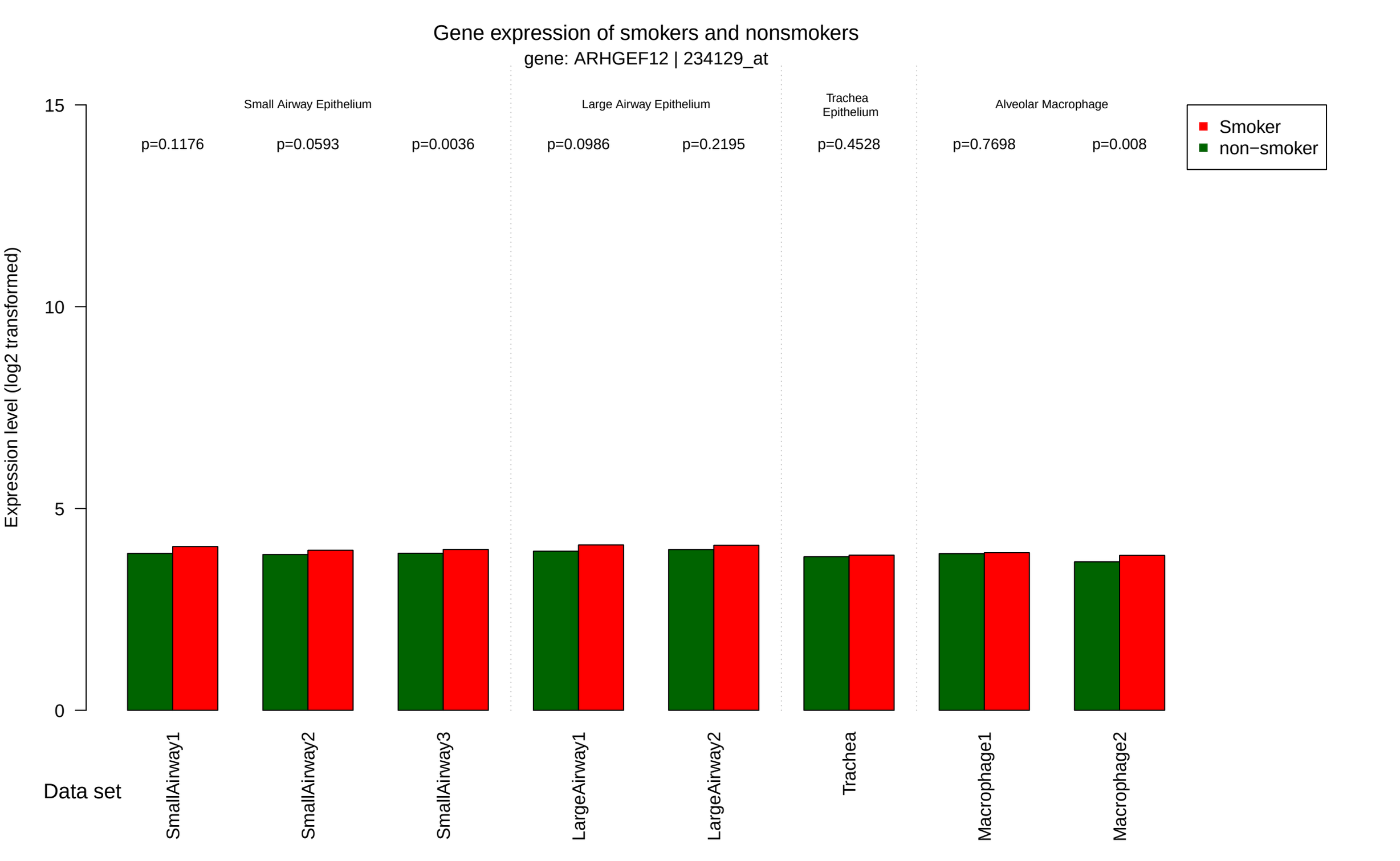
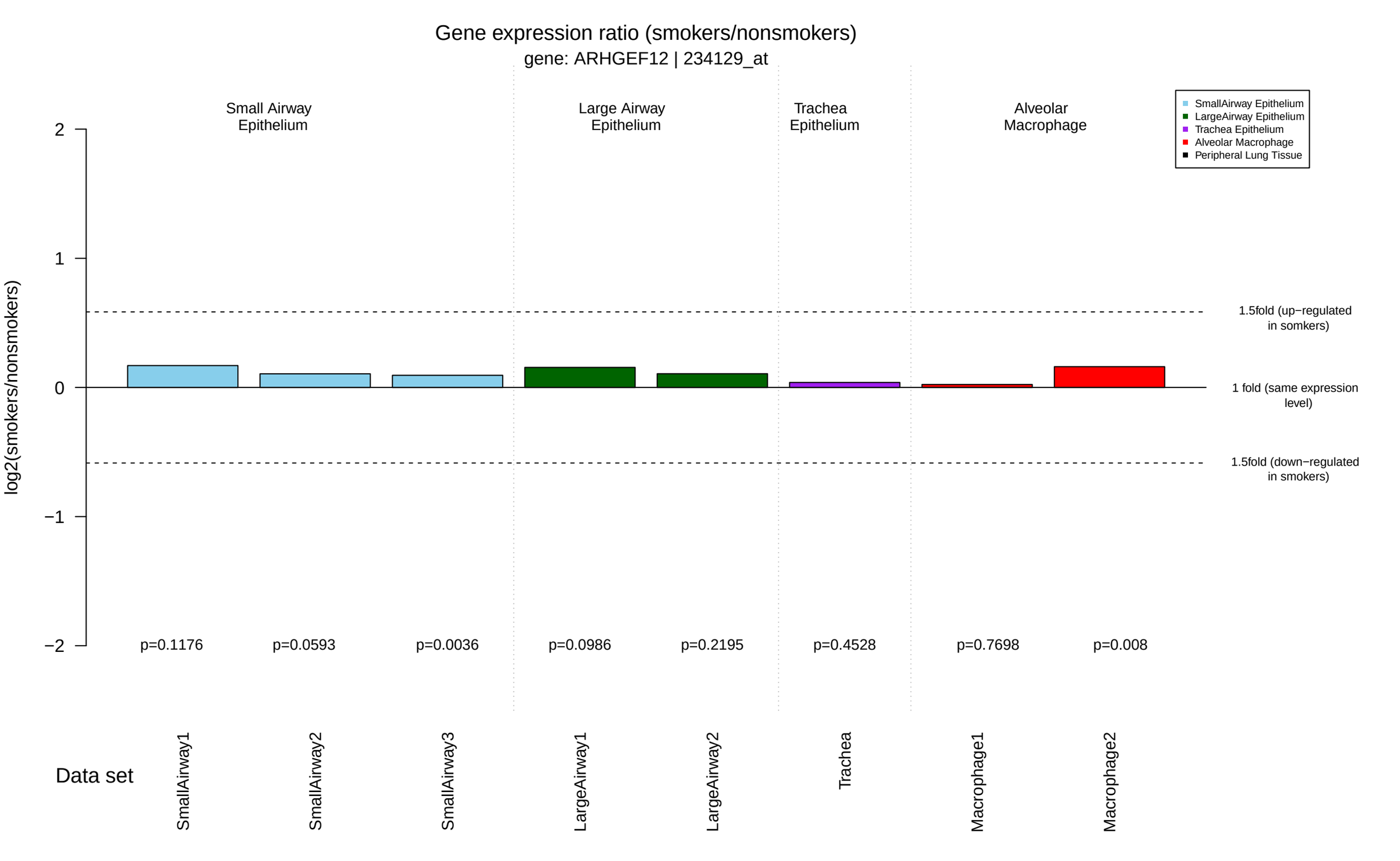
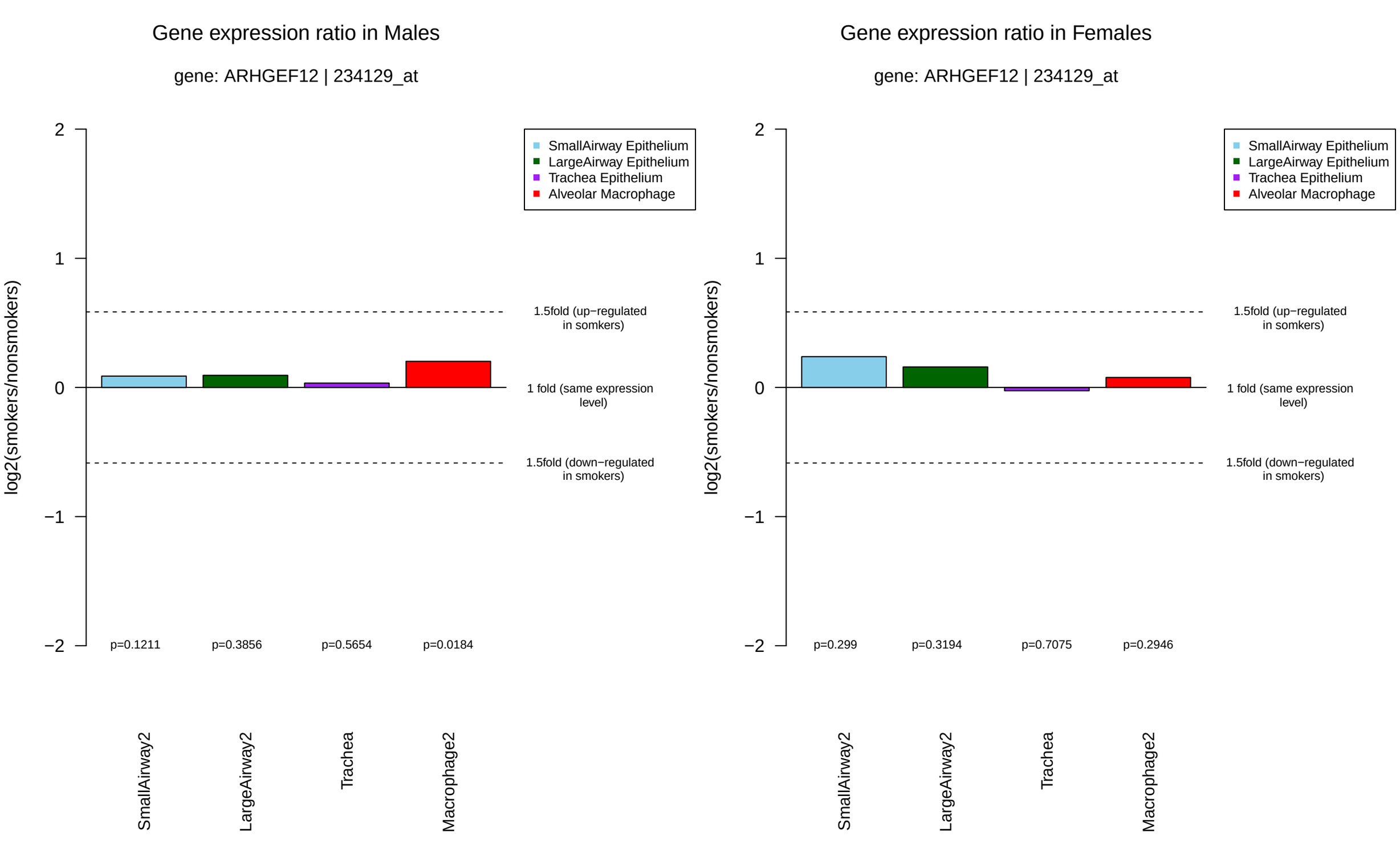
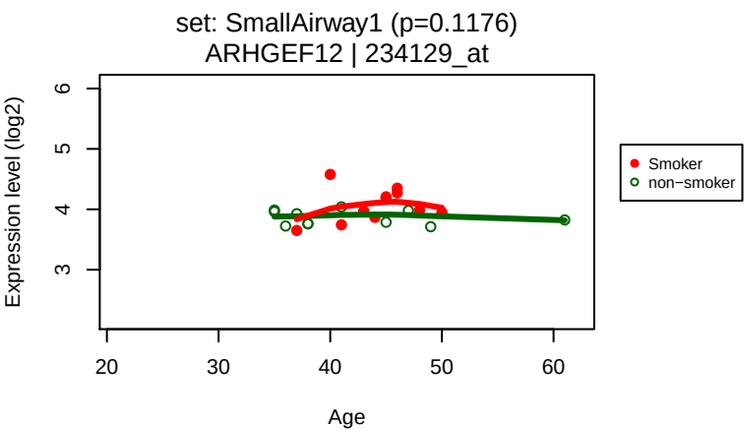
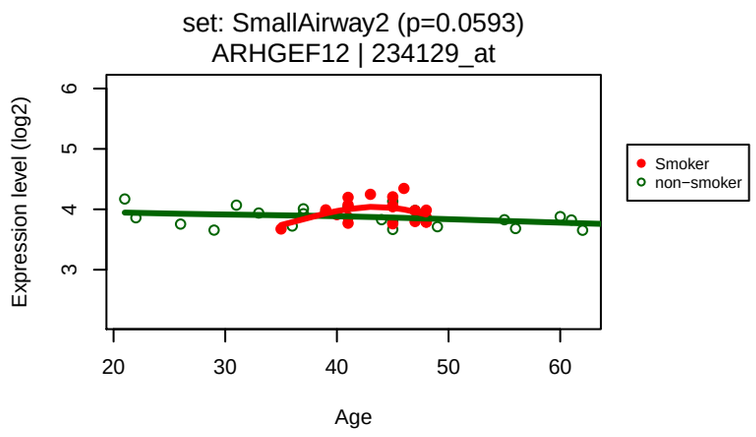
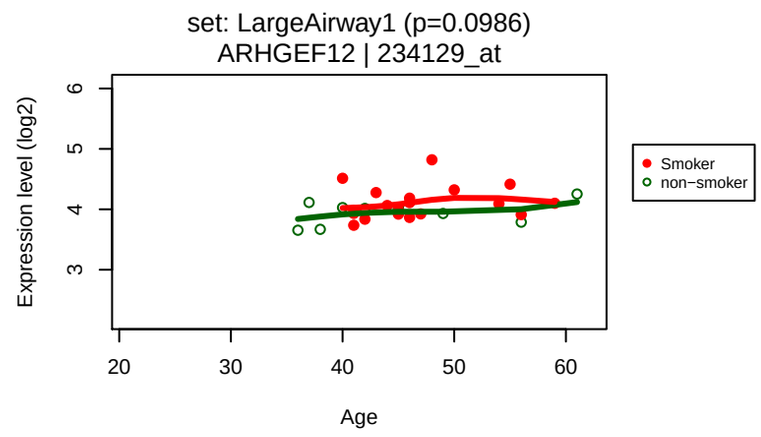

Gene expression of smokers and nonsmokers
gene: ARHGEF12 | 234129_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1176
p=0.0593
p=0.0036
p=0.0986
p=0.2195
p=0.4528
p=0.7698
p=0.008
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

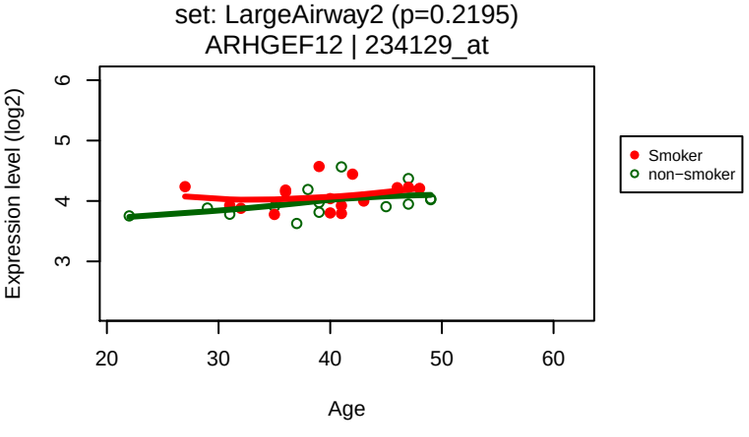

Gene expression ratio (smokers/nonsmokers)
gene: ARHGEF12 | 234129_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1176
p=0.0593
p=0.0036
p=0.0986
p=0.2195
p=0.4528
p=0.7698
p=0.008
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

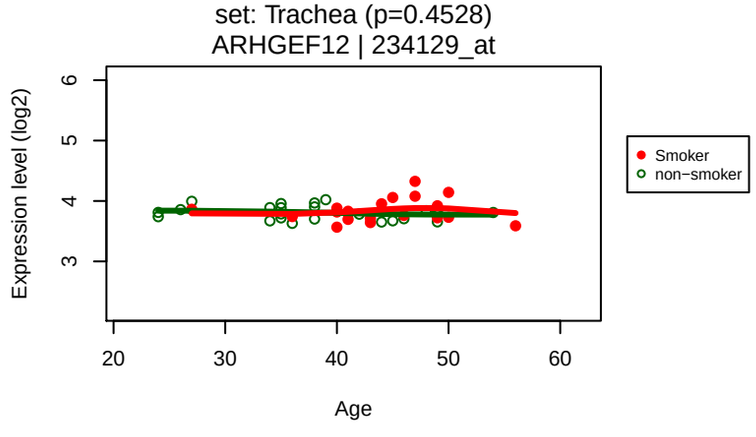

Gene expression ratio in Males
gene: ARHGEF12 | 234129_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1211
p=0.3856
p=0.5654
p=0.0184
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGEF12 | 234129_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.299
p=0.3194
p=0.7075
p=0.2946
SmallAirway2
LargeAirway2
Trachea
Macrophage2

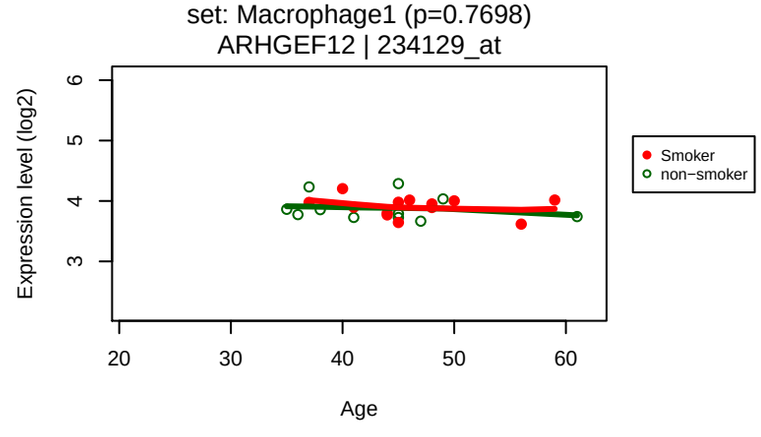

set: SmallAirway1 (p=0.1176)
ARHGEF12 | 234129_at
Expression level (log2)
Age
Smoker
non−smoker

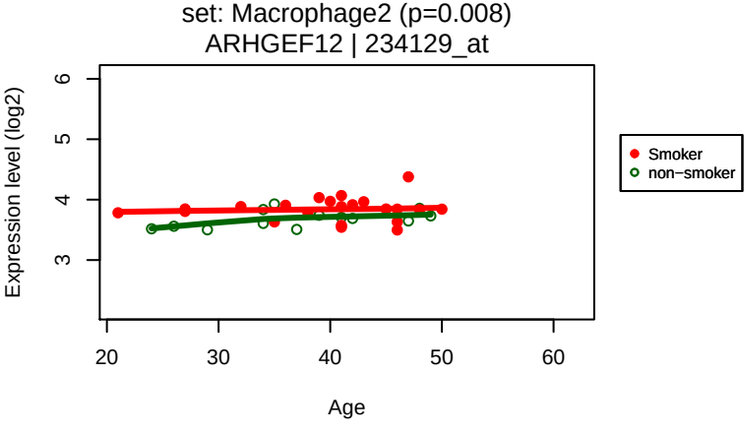

set: SmallAirway2 (p=0.0593)
ARHGEF12 | 234129_at
Expression level (log2)
Age
Smoker
non−smoker

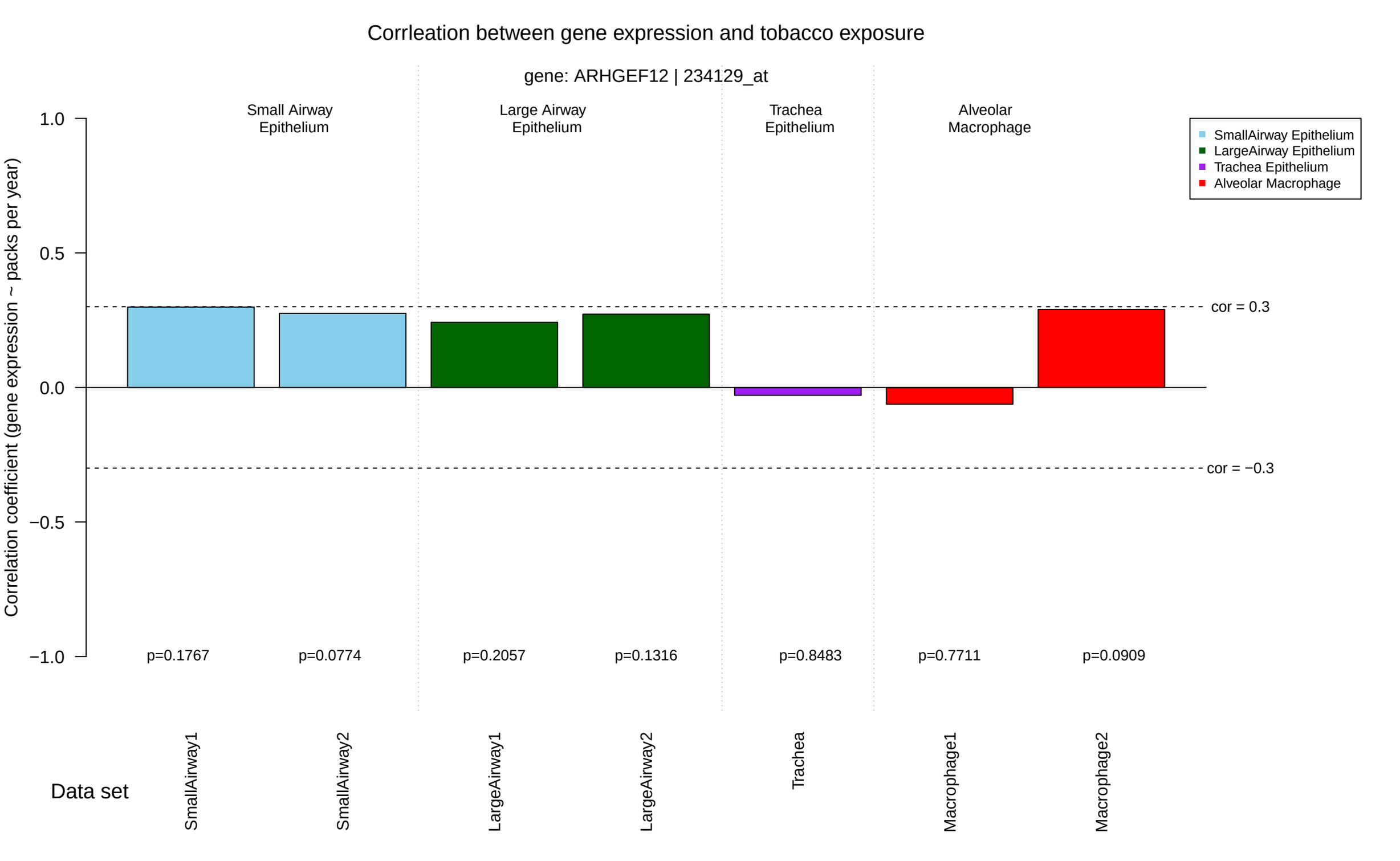

set: LargeAirway1 (p=0.0986)
ARHGEF12 | 234129_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2195)
ARHGEF12 | 234129_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4528)
ARHGEF12 | 234129_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7698)
ARHGEF12 | 234129_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.008)
ARHGEF12 | 234129_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGEF12 | 234129_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1767
p=0.0774
p=0.2057
p=0.1316
p=0.8483
p=0.7711
p=0.0909
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2